System of Units & SI Units

1. System of Units

- A system of units is a set of standard units used to measure physical quantities.
- It provides a common language for scientists, engineers and researchers across the world.
- **Types:**
  1. CGS – centimetre, gram, second
  2. FPS – foot, pound, second
  3. MKS – metre, kilogram, second
  4. SI – International System of Units (most widely used today)

2. SI Units

- SI (Système International d'Unités) is the modern form of MKS system.
- It is based on 7 base units, 2 supplementary units and several derived units.

(a) Base Units (7)

Physical Quantity	Unit Name	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

(b) Supplementary Units (2)

- Plane angle → radian (rad)
- Solid angle → steradian (sr)

(c) Derived Units (Examples)

- Force → newton (N) = $kg\,m\,s^{-2}$
- Pressure → pascal (Pa) = $N\,m^{-2}$ = $kg\,m^{-1}\,s^{-2}$
- Energy, Work, Heat → joule (J) = $N\,m$ = $kg\,m^2\,s^{-2}$
- Power → watt (W) = $J\,s^{-1}$ = $kg\,m^2\,s^{-3}$
- Electric charge → coulomb (C) = $A\,s$
- Electric potential → volt (V) = $W\,A^{-1}$ = $kg\,m^2\,s^{-3}\,A^{-1}$
- Resistance → ohm (Ω) = $V\,A^{-1}$ = $kg\,m^2\,s^{-3}\,A^{-2}$
- Magnetic flux → weber (Wb) = $V\,s$ = $kg\,m^2\,s^{-2}\,A^{-1}$
- Magnetic flux density → tesla (T) = $Wb\,m^{-2}$ = $kg\,s^{-2}\,A^{-1}$
- Frequency → hertz (Hz) = s^{-1}

Dimensional Formulae (Important)

- Force : $[M\,L\,T^{-2}]$
- Pressure : $[M\,L^{-1}\,T^{-2}]$
- Energy/Work : $[M\,L^2\,T^{-2}]$
- Power : $[M\,L^2\,T^{-3}]$
- Electric field : $[M\,L\,T^{-3}\,A^{-1}]$
- Magnetic field : $[M\,T^{-2}\,A^{-1}]$

> Remember: Dimension is the power of base units in a physical quantity.

3. Important Relations & Conversions

(a) Prefixes (Common)

Prefix	Symbol	Factor
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
hecto	h	10^{2}
deca	da	10^{1}

Prefix	Symbol	Factor
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

(b) Some Quick Conversions

- 1 km = 10^3 m
- 1 cm = 10^{-2} m
- 1 mm = 10^{-3} m
- 1 μm = 10^{-6} m
- 1 nm = 10^{-9} m
- 1 Å = 10^{-10} m
- 1 h = 3600 s
- 1 min = 60 s
- 1 day = 86400 s
- 1 u (atomic mass unit) = 1.66×10^{-27} kg

(c) Relations

- 1 N = 10^5 dyne
- 1 erg = 10^{-7} J
- 1 eV = 1.6×10^{-19} J
- 1 cal = 4.184 J

> 1 atm = 1.013×10^5 Pa
> = 760 mm Hg
> = 1.013 bar

4. Best JEE Material Notes

Key Points for JEE

→ Be clear with dimensions of all physical quantities.
→ Learn and remember SI base units and prefixes.
→ Practice dimensional analysis for formula checking and unit conversion.
→ Keep common conversions (like eV, erg, cal, atm, etc.) well memorised.
→ In numerical questions, always write the final answer with proper unit.
→ For JEE Advanced, dimensional reasoning is often used to eliminate options or find relations between physical quantities.

Important Formulae (Quick Recall)

- Velocity : $[L\,T^{-1}]$
- Acceleration : $[L\,T^{-2}]$
- Momentum : $[M\,L\,T^{-1}]$
- Angular momentum : $[M\,L^2\,T^{-1}]$
- Torque : $[M\,L^2\,T^{-2}]$
- Gravitational constant (G) : $[M^{-1}\,L^3\,T^{-2}]$

5. Common Mistakes (Avoid These)

★ Confusing mass and weight (weight is a force).
★ Writing wrong unit in final answer (e.g. N instead of J).
★ Forgetting to convert units before calculation.
★ Ignoring prefixes (k, m, μ, etc.) in numerical problems.
★ Mistaking angle (degree) with radian (in SI, angle is dimensionless).
★ Not checking dimensions in option elimination questions.

Right Units + Right Dimensions = Right Answer

Dimensions and Dimensional Formulae

1. Dimensions – Basic Concept

- Dimensions represent the fundamental nature of a physical quantity in terms of base quantities.
- It tells us how a physical quantity changes when the units of base quantities are changed.
- Two quantities can be added or subtracted only if they have the same dimensions.
- If dimensions are same, then their ratio is dimensionless (pure number).

Dimension of a quantity = $[M^a L^b T^c]$

where,

- M → mass
- L → length
- T → time

For most JEE problems, we use only M, L, T (rarely Θ, I, N etc.)

2. Base and Derived Quantities

(a) Base Quantities (7)

Physical Quantity	Symbol	Dimension
Length	L	[L]
Mass	M	[M]
Time	T	[T]
Electric current	I	[I]
Thermodynamic temperature	Θ	[Θ]
Amount of substance	N	[N]
Luminous intensity	J	[J]

(b) Derived Quantities

- All other quantities can be expressed in terms of base quantities.
- Their dimensions are obtained using dimensional formulae.

3. Rules of Dimensional Formulae

(1) Dimensions of a product = Product of dimensions

$$[AB] = [A][B]$$

(2) Dimensions of a quotient = Quotient of dimensions

$$\left[\frac{A}{B}\right] = \frac{[A]}{[B]}$$

(3) Dimensions of a power = Power of dimensions

$$[A^n] = [A]^n$$

(4) Dimensions of a function = Same as argument

eg. $\sin\theta$, $\cos\theta$, e^x, $\ln x$ → dimensionless

(θ must be dimensionless)

(5) Dimensions of a constant = Dimensionless

eg. 1, 2, π, e, etc.

Important: If the dimensions on both sides of an equation are different, the equation is incorrect.

4. Dimensional Formulae of Some Important Quantities

Quantity	Symbol	Dimensional Formula
Velocity	v	$[LT^{-1}]$
Acceleration	a	$[LT^{-2}]$
Force	F	$[MLT^{-2}]$
Pressure	P	$[ML^{-1}T^{-2}]$
Work / Energy	W, E	$[ML^2T^{-2}]$
Power	P	$[ML^2T^{-3}]$
Momentum	p	$[MLT^{-1}]$
Angular momentum	L	$[ML^2T^{-1}]$
Torque	τ	$[ML^2T^{-2}]$
Gravitational constant	G	$[M^{-1}L^3T^{-2}]$
Planck's constant	h	$[ML^2T^{-1}]$
Electric charge	q	$[IT]$
Electric potential	V	$[ML^2T^{-3}I^{-1}]$
Resistance	R	$[ML^2T^{-3}I^{-2}]$

5. Dimensional Analysis – Applications

(a) Checking the Correctness of an Equation

→ Dimensions of LHS = Dimensions of RHS

(b) Finding the Form of a Relation

- Assume a general form and use dimensions.
- e.g. If $s \propto v^a t^b$, then $[s] = [L]$, $[v] = [LT^{-1}]$, $[t] = [T]$

$$\Rightarrow [L] = [LT^{-1}]^a [T]^b = [L^a T^{-a+b}]$$

$$\Rightarrow a = 1, \quad b = 0 \Rightarrow s \propto v$$

(c) Finding a Missing Dimension

- Use the dimensional formula of the final quantity and equate powers of M, L, T etc.
- e.g. If $[F] = [M^a L^b T^c]$, then $a = 1$, $b = 1$, $c = -2$

$$\Rightarrow [F] = [MLT^{-2}]$$

(d) Converting Units

- Use dimensional formula and known relations.

e.g. 1 cal = 4.184 J, 1 eV = 1.6×10^{-19} J

1 atm = 1.013×10^5 Pa

6. Common Dimensional Pairs (Useful for JEE)

- Energy and Work : $[ML^2T^{-2}]$
- Impulse and Momentum : $[MLT^{-1}]$
- Pressure and Stress : $[ML^{-1}T^{-2}]$
- Surface Tension : $[MT^{-2}]$ (same as stress)
- Angle (plane angle) : [1] (dimensionless)
- Angular velocity : $[T^{-1}]$
- Angular acceleration : $[T^{-2}]$

Remember: Angular quantities are dimensionless or have $[T^{-1}]$, $[T^{-2}]$ etc. depending on the order.

7. Quick Tips for JEE

- ✓ Always write dimensions in square brackets [ ].
- ✓ Use only M, L, T (unless the question involves electricity, temperature, etc.).
- ✓ Check dimension of both sides in formula derivations.
- ✓ For dimensionless quantities, result must be a pure number (or angle in radians).
- ✓ In multiple choice questions, eliminate options using dimensions first.

Key Takeaways

- → Dimensions guide you to the correct formula.
- → They help in checking the plausibility of equations.
- → They are very useful in conversions and in finding missing quantities.
- → In JEE, dimensional analysis is frequently asked in both theory and numerical questions.
- → **Dimension = Power of M, L, T (and sometimes others).**

Dimensional Analysis & Its Applications

1. What is Dimensional Analysis ?

- It is a method of checking the dimensional consistency of physical equations and deducing relations between physical quantities.
- It is based on the principle that in any correct physical equation, the dimensions of all terms must be the same.
- It helps in deriving formulae, finding missing constants and checking the correctness of an equation.

Basic Principle :

> "If a physical quantity is added, subtracted, multiplied or divided, the resultant must have the same dimensions as the original quantity."

2. Rules of Dimensional Analysis

1. Addition / Subtraction : Only same dimensions can be added or subtracted.
   e.g. $[A] = [B] \Rightarrow A + B$ is possible.
2. Multiplication : $[AB] = [A][B]$
3. Division : $\left[\frac{A}{B}\right] = \left[\frac{A}{B}\right]$
4. Power : $[A^n] = [A]^n$
5. Root : $[\sqrt{A}] = [A]^{1/2}$
6. Function : If $y = f(x)$, then $[y] = [x]$
   (for example, trigonometric, logarithmic, exponential functions are dimensionless)

3. Dimensional Formulae of Some Important Quantities

Physical Quantity	Symbol	Dimension
Length	L	$[L]$
Mass	M	$[M]$
Time	T	$[T]$
Force	F	$[M L T^{-2}]$
Pressure	P	$[M L^{-1} T^{-2}]$
Energy / Work	E, W	$[M L^2 T^{-2}]$
Power	P	$[M L^2 T^{-3}]$
Momentum	P	$[M L T^{-1}]$
Angular momentum	L	$[M L^2 T^{-1}]$
Gravitational constant	G	$[M^{-1} L^3 T^{-2}]$
Planck's constant	h	$[M L^2 T^{-1}]$
Electric charge	q	$[I T]$
Electric potential	V	$[M L^2 T^{-3} I^{-1}]$
Resistance	R	$[M L^2 T^{-3} I^{-2}]$

4. Applications of Dimensional Analysis

(a) Deriving Formulae :

- Assume a relation $y = k x^a y^b z^c \ldots$
- By equating dimensions, find the powers and constant.

e.g. $v = k\sqrt{gl} \Rightarrow [v] = [\sqrt{gl}]$

$\Rightarrow [LT^{-1}] = [LT^{-2}]^{1/2}[L]^{1/2} = [LT^{-1}]$ ✓

(b) Finding Unknown Dimensions :

- If a physical quantity depends on unknown constants, its dimension can be found.

e.g. $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow [T] = \left[\left(\frac{L}{LT^{-2}}\right)^{1/2}\right] = [T]$ ✓

(c) Checking the Correctness of an Equation :

- All terms must have the same dimensions.

e.g. In $s = ut + \frac{1}{2}at^2$,

$[s] = [ut] = [at^2] = [L]$ ✓

(d) Finding a Missing Dimension / Constant :

- Use dimensional formula to find the unknown quantity or value of a constant.

e.g. If $F = G\frac{m^a R^b}{r^2}$, then $a = 1, b = 2, c = 2$

$\Rightarrow G = [M^{-1} L^3 T^{-2}]$

5. Important Dimensional Applications (JEE Focus)

1. Kepler's Third Law : $T^2 \propto r^3 \Rightarrow T^2 = kr^3$
   $\Rightarrow [k] = [T^2 L^{-3}]$
2. Simple Pendulum : $T = 2\pi\sqrt{\frac{l}{g}}$
   $\Rightarrow [T] = \left[\left(\frac{L}{LT^{-2}}\right)^{1/2}\right] = [T]$ ✓
3. Projectile Motion : $R = \frac{u^2 \sin 2\theta}{g}$.
   $\Rightarrow [R] = [L]$ ✓
4. Drag Force : $F = \frac{1}{2}\rho v^2 A C$
   $\Rightarrow [F] = [MLT^{-2}]$ ✓
5. Gravitational Potential : $V = -\frac{GM}{r}$
   $\Rightarrow [V] = [L^2 T^{-2}]$ ✓

6. Common Mistakes (Avoid These)

- Adding or subtracting quantities of different dimensions.
- Ignoring the dimensions of constants (unless stated).
- Wrongly handling powers and roots.
- Not converting all quantities to the same system of units.
- Mistaking dimensionless quantities for having dimensions.

7. Dimensional Analysis in JEE

- Used in numerical value problems (finding constants).
- Used in verifying equations and relations.
- Helps in checking options in MCQs (elimination).
- Frequently asked in Modern Physics, Mechanics, Fluid Mechanics and Electrodynamics.

Key Takeaways

★ Dimensions help in formulating and checking physical equations.
★ The dimension of a physical constant is not zero.
★ Always convert to the same system of units.
★ Dimensional analysis simplifies complex problems in JEE.
★ It is a powerful tool for problem solving and concept clarity.

→ Dimension = Power of M, L, T (and sometimes I)

Significant Figures

1. Introduction

- Significant figures (s.f.) are the correctly known digits in a measured value, including the first uncertain digit.
- They indicate the precision of a measurement.
- They are used in calculations, results and expressing the accuracy of a value.

Example :

- 2.36 → 3 s.f.
- 0.00750 → 3 s.f.
- 1200 → 2 s.f. (if not specified, trailing zeros without decimal point are not significant)

2. Rules for Identifying Significant Figures

(a) Non-zero Digits

- All non-zero digits are significant.

Ex : 3.425 → 4 s.f.
7.8 → 2 s.f.
560.2 → 4 s.f.

(b) Zeros Between Non-zero Digits

- Zeros between non-zero digits are significant.

Ex : 1.005 → 4 s.f.
6.03 → 3 s.f.
4.200 → 4 s.f.

(c) Leading Zeros (at the beginning)

- Leading zeros are NOT significant.

Ex : 0.00456 → 3 s.f.
0.072 → 2 s.f.
0.00032 → 2 s.f.

(d) Trailing Zeros (at the end)

- 1. If there is a decimal point, trailing zeros are significant.

Ex : 2.30 → 3 s.f.
5.600 → 4 s.f.

- 2. If there is no decimal point, trailing zeros are NOT significant (unless indicated by a bar or scientific notation).

Ex : 1200 → 2 s.f. (not 4)
45000 → 2 s.f. (not 5)
0.0500 → 3 s.f.

(e) Exact Numbers

- Counted or defined numbers have infinite significant figures.

Ex : 1 dozen = 12 (exact)
1 m = 100 cm (exact)

3. Significant Figures in Scientific Notation

- In a number in the form $a \times 10^n$, only the digits in "a" are significant.

Ex : 6.23×10^4 → 3 s.f.
0.000415×10^3 → 3 s.f.
9×10^{-5} → 1 s.f.

The power of 10 does not affect the number of significant figures.

4. Operations with Significant Figures

(a) Addition and Subtraction

- Result should have the same number of decimal places as the least precise measurement.
- **Example** : 3.426 + 1.2 = 4.626 → 4.6 (2 d.p.)
  5.80 − 2.1 = 3.7 (1 d.p.)

(b) Multiplication and Division

- Result should have the same number of significant figures as the least precise value.
- **Example** : 2.34 × 3.1 = 7.254 → 7.3 (2 s.f.)
  12.5 ÷ 4.0 = 3.125 → 3.1 (2 s.f.)

(c) Mixed Operations

- Follow the order: (1) Brackets (2) × and ÷ (3) + and −
- Keep s.f. at each step.
- **Example** : (2.50 × 3.0) + 1.2
  = 7.5 + 1.2 = 8.7 (2 d.p. and 2 d.p.)
  (3.2 × 4.5) ÷ 2.0
  = 14.4 ÷ 2.0 = 7.2 (2 s.f.)

5. Rounding Off Rules

- If the next digit is 0, 1, 2, 3, or 4 → keep the digit unchanged.
- If the next digit is 5, 6, 7, 8, or 9 → increase the preceding digit by 1.

Examples :

- 2.346 (3 s.f.) → 2.35
- 5.678 (3 s.f.) → 5.68
- 1.995 (3 s.f.) → 2.00
- 0.0496 (2 s.f.) → 0.050

In JEE, always write the final answer with the correct number of significant figures / decimal places.

6. Important Points for JEE

- In numerical problems, always maintain the correct number of significant figures in intermediate steps.
- When using constants (g, R, etc.), use the value given in the question with its s.f.
- In physics, while taking measurements, the s.f. depends on the instrument's least count.
- For logarithms and antilogarithms, the number of decimal places in the mantissa matters.
- In a final answer, never add unnecessary zeros.

Quick Revision Table

Case	Significant Figures
Non-zero digits	Always significant
Zeros between non-zero	Significant
Leading zeros	Not significant
Trailing zeros (with decimal)	Significant
Trailing zeros (without decimal)	Not significant
Exact numbers	Infinite

★ **Remember** : Precision is not accuracy, but it gives the limit of accuracy.

Errors in Measurement

1. Introduction

- No measurement is exact. Every measurement contains some error.
- Errors arise due to limitations of instruments, observation, environment, method, etc.
- These errors are unavoidable, but can be reduced and estimated.

2. Types of Errors

(a) Systematic Errors

- Have a definite pattern and same sign.
- Caused by faulty instruments, method, calibration, neglecting standards, etc.
- Can be eliminated or corrected (if known).
- **Example**: Zero error in a vernier caliper.

(b) Random Errors

- Arise due to uncontrollable variations.
- Have no definite pattern; can be positive or negative.
- Reduced by taking repeated measurements and finding average.
- **Example**: Reading a scale slightly different each time.

(c) Gross Errors

- Caused by human mistakes (wrong reading, calculation, transcription, etc.).
- Can be avoided by careful observation and practice.
- **Example**: Taking 5.2 cm instead of 4.2 cm.

3. Absolute and Relative Error

(a) Absolute Error. (Δx)

- $\Delta x = |\text{Measured value} - \text{True value}|$

  (or just the uncertainty in the measurement)

(b) Relative Error

$$\text{Relative error} = \frac{\Delta x}{\text{True value}} = \frac{\Delta x}{x} \quad \text{(approx)}$$

(c) Percentage Error

$$\text{Percentage error} = \left(\frac{\Delta x}{x}\right) \times 100\%$$

Example :

A length is measured as 5.2 cm with an absolute error of 0.1 cm. Find the relative and percentage error.

Soln : $x = 5.2$ cm, $\Delta x = 0.1$ cm

$$\text{Relative error} = \frac{0.1}{5.2} = 0.0192 = 1.92\%$$

$$\text{Percentage error} = 1.92\%$$

4. Types of Uncertainties

- **Absolute uncertainty** : $\pm \Delta x$
- **Relative uncertainity** : $\frac{\Delta x}{x}$
- **Percentage uncertainty** : $\frac{\Delta x}{x} \times 100\%$

In JEE, uncertainty is mostly asked in % form or in terms of significant figures.

5. Propagation of Errors

(a) For Addition / Subtraction

$$\boxed{\Delta z = \Delta x + \Delta y} \quad \text{(absolute error)}$$

Example : $z = x + y = 5.2 + 3.1 = 8.3$

$\Delta z = 0.1 + 0.2 = 0.3$

(b) For Multiplication / Division

$$\boxed{\frac{\Delta z}{z} = \frac{\Delta x}{x} + \frac{\Delta y}{y}} \quad \text{(for product or quotient)}$$

Example : $z = x \times y$ $\quad x = 5.0 \pm 0.1$, $\quad y = 4.0 \pm 0.2$

$$\frac{\Delta z}{z} = \frac{0.1}{5.0} + \frac{0.2}{4.0} = 0.02 + 0.05 = 0.07$$

$\therefore \Delta z = 7\%$ of z

(c) For Powers and Roots

$$\boxed{\frac{\Delta z}{z} = n\frac{\Delta x}{x}} \quad (\text{if } z = x^n)$$

Example : $z = x^2$, $\quad x = 3.0 \pm 0.1$

$$\frac{\Delta z}{z} = \frac{2 \times 0.1}{3.0} = 0.0667 = 6.67\%$$

6. Some Important Points for JEE

- In addition and subtraction, absolute errors add.
- In multiplication and division, relative errors add.
- For a constant (e.g. π, g, etc.), error is negligible.
- Percentage error is useful for comparison.
- If a quantity is measured indirectly, use propagation of errors.
- In JEE, numerical problems often involve error in constants, instrument least count, or multiple steps of calculation.

7. Common Mistakes (Avoid These)

- Treating absolute error and relative error as the same.
- Ignoring sign of error in addition/subtraction.
- Forgetting to convert percentage into decimal form.
- Not considering significant figures while writing final answer.
- Confusing random error with systematic error.

Quick Revision Table

Quantity	Formula / Rule
Absolute error	$\Delta x = \lvert x_{meas} - x_{true} \rvert$
Relative error	$\frac{\Delta x}{x}$
Percentage error	$\frac{\Delta x}{x} \times 100\%$
Addition/Subtraction	$\Delta z = \Delta x + \Delta y$
Multiplication/Division	$\frac{\Delta z}{z} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$
Power ($z = x^n$)	$\frac{\Delta z}{z} = n\frac{\Delta x}{x}$

★ **Remember** : Error can never be zero, but it can be minimized. More measurements → less random error (within limit).